

SUPPLEMENTARY RESULTS: GOPT

This document contains the model-specific component and loss-function ablations for GOPT, accompanying *Beyond the Majority Bin: A Methodological Framework for Balanced Automatic Pronunciation Assessment*. The cross-model comparisons and experimental reproducibility settings are reported in the paper’s supplementary material.

I. READING GUIDE

Baseline denotes the original model and training procedure. The +CV configuration introduces speaker-independent grouped cross-validation using `GroupKFold(n_splits=5)` without shuffling. Both +DB and +MS build on CV: +DB adds Dynamic Balanced MSE, whereas +MS adds Macro-MSE-based checkpoint selection. Complete BOMA combines CV, DB, and MS. In the loss-function ablations, CV and MS are held fixed while conventional MSE, DB-MSE, and Score-Balanced Loss (SB) are compared.

CCC-BS measures distribution-balanced agreement and PCC measures linear association; higher values are preferred for both. Micro-MSE and Macro-MSE are error measures on the unified $[0, 10]$ score scale, so lower values are preferred. Boldface marks the best value in each comparison. These metrics express complementary evaluation objectives; improvements under balanced criteria need not coincide with improvements under sample-weighted criteria.

II. RESULTS

For GOPT, complete BOMA obtains the best CCC-BS on five of the nine scoring aspects, whereas CV+MS is best on the remaining four (Table S1). The Macro-MSE analysis shows the same complementarity: BOMA yields the lowest Macro-MSE on seven aspects and CV+MS on two, while the lowest Micro-MSE is usually obtained by the Baseline (Table S2). PCC instead favors the Baseline or CV+MS on all aspects, with CV tying the Baseline on utterance accuracy and fluency (Table S3). In the controlled loss comparison, DB-MSE is strongest on four CCC-BS aspects, MSE on four, and SB on utterance fluency; DB-MSE gives the lowest Macro-MSE on six of nine aspects (Tables S4 and S5).

TABLE S1
ABLATION STUDY FOR GOPT. CORRELATIONS ARE MEASURED USING CCC-BS.

Configuration	Phone Score	Word Score (CCC-BS)				Utterance Score (CCC-BS)			
	Phoneme ↑	Acc. ↑	Stress ↑	Total ↑	Acc. ↑	Comp. ↑	Fluency ↑	Prosody ↑	Total ↑
Baseline	0.388	0.457	0.117	0.521	0.731	0.025	0.757	0.759	0.776
+ CV	0.412	0.485	0.121	0.531	0.743	0.064	0.778	0.739	0.780
+ DB	0.526	0.585	0.517	0.487	0.703	0.087	0.712	0.734	0.751
+ MS	0.496	0.557	0.168	0.629	0.755	0.115	0.790	0.806	0.831
BOMA	0.597	0.595	0.599	0.528	0.748	0.046	0.796	0.819	0.792

^a The +DB and +MS configurations are evaluated on top of the CV configuration; BOMA combines CV, DB, and MS.

^b Bold values indicate the highest CCC-BS value for each aspect within this ablation.

TABLE S2
MICRO-MSE AND MACRO-MSE ABLATION STUDY FOR GOPT. LOWER VALUES INDICATE BETTER PERFORMANCE.

	Phone		Word Score						Utterance Score									
	Phoneme		Acc.		Stress		Total		Acc.		Comp.		Fluency		Prosody		Total	
Configuration	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓
Baseline	2.117	17.794	2.308	12.973	0.179	9.922	1.748	11.588	1.175	3.231	0.255	27.905	0.861	4.093	0.877	3.838	1.095	3.620
+ CV	2.170	16.817	2.341	12.044	0.180	9.900	1.791	10.948	1.196	2.975	0.245	26.217	0.922	3.701	0.923	4.049	1.115	3.446
+ DB	5.078	8.940	8.534	5.618	1.953	4.082	5.160	7.731	1.459	2.968	0.313	24.541	1.213	5.003	1.072	4.216	1.277	3.945
+ MS	2.474	13.112	2.572	9.571	0.186	9.342	2.024	8.132	1.459	2.460	0.239	23.973	1.190	3.179	1.071	2.903	1.327	2.505
BOMA	7.635	6.664	10.785	5.342	2.655	3.742	6.308	6.341	1.934	2.277	0.260	25.256	1.410	3.139	1.347	2.624	1.737	2.913

^a Micro-MSE (Mic.) and Macro-MSE (Mac.) are reported on the unified $[0, 10]$ scale; lower values (↓) indicate better performance.

^b The +DB and +MS configurations are evaluated on top of CV; BOMA combines CV, DB, and MS.

^c Bold values indicate the lowest error for each metric within the model-specific ablation.

TABLE S3
PCC ABLATION STUDY FOR GOPT. HIGHER VALUES INDICATE BETTER PERFORMANCE.

Configuration	Phone Score	Word Score (PCC)			Utterance Score (PCC)				
	Phoneme ↑	Acc. ↑	Stress ↑	Total ↑	Acc. ↑	Comp. ↑	Fluency ↑	Prosody ↑	Total ↑
Baseline	0.612	0.533	0.291	0.549	0.714	0.155	0.753	0.760	0.742
+ CV	0.606	0.527	0.337	0.540	0.714	0.367	0.753	0.750	0.738
+ DB	0.497	0.435	0.212	0.462	0.692	0.180	0.730	0.733	0.721
+ MS	0.589	0.515	0.346	0.530	0.710	0.441	0.738	0.746	0.739
BOMA	0.482	0.422	0.193	0.437	0.674	0.236	0.728	0.733	0.709

^a The +DB and +MS configurations are evaluated on top of CV; BOMA combines CV, DB, and MS.

^b Bold values indicate the highest PCC value for each aspect within the model-specific ablation.

TABLE S4
LOSS-FUNCTION ABLATION IN TERMS OF CCC-BS FOR GOPT. ALL CONFIGURATIONS USE THE SAME CROSS-VALIDATION AND MACRO-MSE-BASED MODEL-SELECTION PROTOCOL; ONLY THE TRAINING LOSS IS CHANGED.

Loss	Phone Score	Word Score (CCC-BS)			Utterance Score (CCC-BS)				
	Phoneme ↑	Acc. ↑	Stress ↑	Total ↑	Acc. ↑	Comp. ↑	Fluency ↑	Prosody ↑	Total ↑
MSE	0.496	0.557	0.168	0.629	0.755	0.115	0.790	0.806	0.831
DB-MSE	0.597	0.595	0.599	0.528	0.748	0.046	0.796	0.819	0.792
SB	0.461	0.535	0.189	0.602	0.738	0.108	0.819	0.784	0.818

^a MSE corresponds to the CV+MS configuration, i.e., the BOMA validation and model-selection protocol trained with conventional MSE.

^b DB-MSE corresponds to the proposed Dynamic Balanced MSE used in the complete BOMA framework.

^c SB denotes Score-Balanced Loss, which replaces DB-MSE while keeping the remaining protocol unchanged.

^d Bold values indicate the highest CCC-BS for each aspect within the model-specific loss ablation.

TABLE S5
LOSS-FUNCTION ABLATION IN TERMS OF MICRO-MSE AND MACRO-MSE FOR GOPT. ALL CONFIGURATIONS USE THE SAME CROSS-VALIDATION AND MACRO-MSE-BASED MODEL-SELECTION PROTOCOL; ONLY THE TRAINING LOSS IS CHANGED.

Loss	Phone		Word Score						Utterance Score									
	Phoneme		Acc.		Stress		Total		Acc.		Comp.		Fluency		Prosody		Total	
	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓
MSE	2.474	13.112	2.572	9.571	0.186	9.342	2.024	8.132	1.459	2.460	0.239	23.973	1.190	3.179	1.071	2.903	1.327	2.505
DB-MSE	7.635	6.664	10.785	5.342	2.655	3.742	6.308	6.341	1.934	2.277	0.260	25.256	1.410	3.139	1.347	2.624	1.737	2.913
SB	2.487	14.039	2.669	9.866	0.187	8.701	2.028	8.801	1.547	2.544	0.228	24.404	1.161	2.784	1.107	3.227	1.452	2.709

^a MSE corresponds to the CV+MS configuration, i.e., the BOMA validation and model-selection protocol trained with conventional MSE.

^b DB-MSE corresponds to the proposed Dynamic Balanced MSE used in the complete BOMA framework.

^c SB denotes Score-Balanced Loss, which replaces DB-MSE while keeping the remaining protocol unchanged.

^d Micro-MSE (Mic.) and Macro-MSE (Mac.) are reported on the unified [0, 10] scale; lower values (↓) indicate better performance.

^e Bold values indicate the lowest error for each metric within the model-specific loss ablation.